Agent Response Analysis with Context-Aware RAGs

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Abstract This paper introduces an agent-based Retrieval-Augmented Generation (RAG) framework that helps users understand long and intricate documents using specialised agents with domain knowledge for better reasoning. While large language models excel at question answering and summarization, they stumble on niche texts requiring structured breakdowns, or cross-sectional analysis.Traditional RAG adds context to improve accuracy, but it follows a rigid process and does not handle retrieval errors well, which can lead to incorrect or misleading answers.We observed consistent improvements in factual reliability and response quality across the Legal (NDA) domain, where lower hallucination rates and higher groundedness and BLEU scores indicate better performance. For NDA documents, incorporating miscellaneous sources improves BLEU from 0.02 to 0.10, while groundedness drops from 0.87 to 0.81 and hallucination rates rise from 0.13 to 0.19, pointing to a trade-off between surface-level similarity and factual accuracy.

Keywords Retrieval Augmented Generation, RAG, Agents, LLM, Large Language Models

I. Introduction

These days, a lot of information lives online. Much of it comes in the form of long documents and PDFs that aren't neatly structured. This is especially common in areas like law, finance, education, and scientific research, where people rely on things like academic papers, policy reports, and legal records. Storing and sharing files digitally is much easier now. But it also makes these documents harder to understand. Important details are often scattered across different sections instead of being clearly organised in one place neatly [7][8][9][10][12][13].

When it comes to tasks like answering questions and summarisation, Large Language Models (LLMs) excel. They typically perform well in everyday situations. But problems arise when they are applied to intricate or high-risk documents. The majority of generic question-answering systems rely on pretrained knowledge, which is fixed, difficult to confirm, and not always directly related to the document that is being analysed. As a result, the model provides answers that seem confident but contain errors and omit crucial information. In areas like finance, engineering and law, these small mistakes can lead to serious consequences [2][7][9][10][12][14].

Retrieval-Augmented Generation (RAG) was introduced to handle these problems by first finding relevant documents and then using them to generate answers [2][8]. This helps improve factual accuracy, but most RAG systems still follow a fixed process. When something goes wrong during retrieval, the AI still uses the incorrect or incomplete information to respond because they typically assume that the information retrieved is accurate and sufficient. It doesn't stop or correct itself, which can result in answers that are false or deceptive.

Current QA and RAG systems also have the problem of treating all documents equally, which isn't very practical. Different types of reasoning are required for different documents. For example, legal documents require clause-by-clause analysis, financial documents involve numbers and rules, and academic papers need an understanding of methods and results. Research from a variety of fields demonstrates that retrieval and reasoning performance can vary significantly based on the type of document, which highlights the limitations of using a single approach for every type of document[4][8][9][10].

Recent research has demonstrated that evaluating RAG systems solely based on how accurate their responses seem is insufficient. Models frequently respond with high confidence, which causes hallucinations, even if the information retrieved is weak or inaccurate. Because of this behaviour, merely verifying the final response does not accurately represent the system's level of reliability. Researchers have begun concentrating on techniques that can identify such discrepancies and update responses using more outside data in order to address this problem. Some methods let the model respond first, then confirm that response with more data, making adjustments if the outcome is not correct. Based on this idea, our work evaluates model outputs not only using conventional evaluation measures, but also through additional measures that assess retrieval quality and groundedness [5][12][13][14].

Based on these ideas, we use a multi-agent system to understand documents. Rather than reading everything at once, the system splits the work into smaller tasks that are handled by specialised agents. These agents work together, sharing their understanding and looking up additional sources whenever something needs clarification. By taking a domain-aware approach, the system produces insights that are easier to understand and grounded in the original documents.

II. Methodology

A. Agentic System

When we query a large language model for specific information, the LLM is trained on billions and millions of pieces of data. So, for the LLM, it becomes difficult to answer the query for a specific question. That is why it attempts to provide a generalised answer. Even for data scientists who work on these LLM architectures, it is more advantageous if the model provides generalised answers. Now, because of this, asking a specific context-related query and expecting such context-aware responses becomes difficult.

We need to build special, dedicated ways to handle information for each unique area of knowledge, just like we talked about at the start. This project is all about creating and using smart agents that really understand their subject. What makes these agents special is that they use a core system called Retrieval-Augmented Generation (RAG), which helps them find the right information to back up their answers. They also come with a set of helpful tools customised for their context.

Now, understanding how the agent works, an agent is basically an LLM that is aware of its environment and queries the tools in the environment. So when we query an LLM, what happens is it queries back to the vectors on which the LLM is trained, matches the distances of vectors and produces the response. Whereas when using an agent, the agent uses its own reasoning to analyse the tasks, it makes it possible to rethink about its actions and then give a response, now these responses also act as observations to the next responses, hence, the hallucination of the LLM model is reduced.[15]

When multiple agents are employed, they are designed to replicate human behaviours such as decision-making, response generation, human-like search capabilities, and reasoning processes. This creates an iterative agentic flow, where each agent continuously interacts and refines its outputs until a satisfactory response is generated. This methodology enhances the quality and relevance of the answers produced.

When we leverage the RAG architecture, we effectively engage with the user's knowledge base by creating robust vector embeddings. This process allows us to search for similarities within the knowledge base efficiently. We then provide this context to the LLM for reasoning, enabling us to confidently analyse the outputs generate.

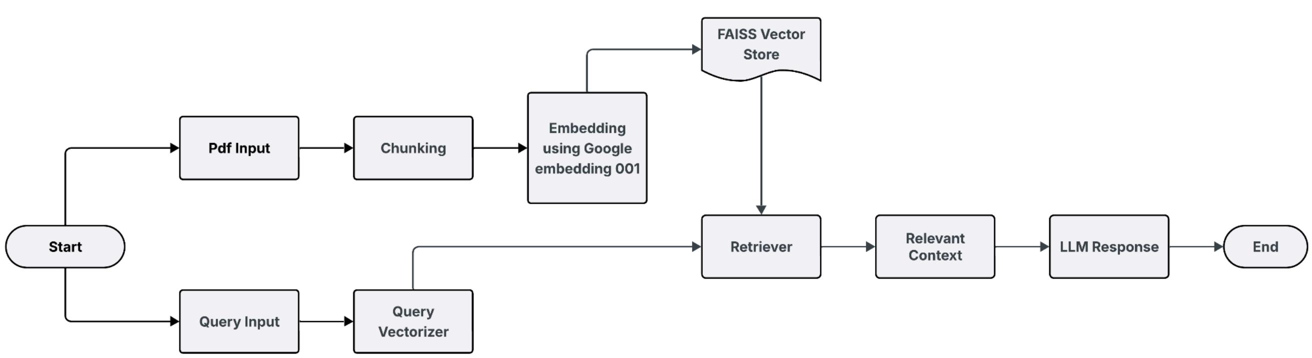

Figure 1. System Architecture Diagram

B. Pipelined structure of RAG

This is the complete RAG architecture, which enables the integration of different stages, including ingestion, chunking, indexing, embedding of queries, retrieval, contextual prompting, and generation. Though the basic implementation of the RAG model lacks complex procedures like routing, re-ranking, and iterative methods, the model has been able to deliver question-and-answer functionality on the PDF content.

A long text is broken down into overlapping shorter segments to preserve context integrity across segment boundaries by using a sliding window technique. This technique starts at position 0 with a window of 600 characters and moves by 520 characters (600 - 80 overlap) till it has processed the whole text. The fixed-size segments with 80 overlaps handle the issue of losing context at boundary positions in embedding-based search, where context integrity has high priority.

A list of chunks of text is represented as dense vectors using the embedding model from Google embedding-001. Chunks are embedded one by one through a loop, gathering embeddings into a numpy array. L2normalisation is performed through FAISS for all vectors for ease of calculation of the cosine similarity of vectors as inner products. The resulting output is a float32 normalised array.

The storage of FAISS is initialised through the creation of an inner product index with dimensions corresponding to the embeddings. This step entails adding all the pre-computed embeddings into the index while maintaining the text chunks that can be retrieved through the index positions.

A similarity search is carried out by passing a normalised query embedding into the search function offered by FAISS. A search returns raw similarities and their corresponding indices, which are then mapped back into original chunks from a list comprehension by using the indices. A search always returns precisely k indices, independent of score thresholds.

The end-to-end query retrieval wrapper involves embedding the query using the same embedding model and normalisation procedure as was employed on the chunks of the documents. The procedure calls a search method of the vector store and then uses a procedure to remove any duplicated chunks while maintaining the order using a slice method for an exact k unique results. All this ensures that the results of context are attained from semantically similar chunks of documents.

Context-aware text generation is aided by the use of the Gemini model (gemini-2.5-flash). A context-aware prompt template is chosen based on the context by incorporating the retrieved context as well as the query. Context-related temperatures are then set according to the type of context, such as 0.3 for finance and 0.5 for legal-related tasks, to avoid hallucinations. Higher temperatures, ranging from 0.7 to 0.9, are used for artistic-related tasks, along with a constant top p of 0.9 for sampling. Additional whitespace characters are removed from the raw model output.

The master orchestrator takes all the components of the system together, performing the loading of the PDF, the chunking of the text, embedding of the chunks, creation of the vector store, retrieval of the top ten chunks, and final use of the top five for contextual input, finally resulting in the generation of a context-aware answer. If a ground truth answer is provided, it allows evaluation metrics to be calculated, but it always provides the answer as well as a dictionary of metrics.

III. LITERATURE SURVEY

Retrieval-Augmented Generation (RAG) has been shown to be a promising paradigm for improving the fact grounding of large language models by leveraging external knowledge sources. Various strategies, including dense retrieval pipelines through to domain-specific customisations and corrective methods, have been introduced in the last couple of years. Since we are undertaking a systematic review of the literature, we can now distill out common trends, innovations, and challenges. The table below provides a framework to compare key works in terms of the forum, techniques, findings, drawbacks, and future direction, enabling us to place our proposed system in the light of the dynamic trend in the literature.

Sr. No.	Journals/Conferences/ Year/Author	Methodology	Results	Key Limitations	Future Scope
[1]	arXiv, 2023 – Wu et al.	Proposed AutoGen, a multi-agent framework enabling next-gen LLM	Demonstrated improved task decomposition, coordination, and autonomy	Coordination overhead, error propagation across agents, and reliance on curated prompts.	More efficient multi-agent orchestration, integration with human-in-the-loop systems, and broader enterprise deployment.

Sr. No.	Journals/Conferences/ Year/Author	Methodology	Results	Key Limitations	Future Scope
		applications via conversational collaboration among agents.	in complex workflows		
[2]	NeurIPS, 2020 – Lewis et al.	Introduced Retrieval-Augmented Generation (RAG) by combining large language models with external knowledge sources.	Improved factual grounding and QA accuracy compared to pure generative models.	Assumes retrieved documents are always relevant.	Adaptive retrieval and error detection mechanisms.
[3]	EMNLP,2020 – Karpukhin et al.	Proposed Dense Passage Retrieval (DPR) for open-domain question answering	High recall and QA accuracy	Performance drops when the domain changes	Domain adaptations hybrid retrieval
[4]	ICLR, 2021 – Thakur et al.	Provided a multi-domain benchmark (BEIR) for evaluating retrievers	Zero-shot retrieval effectiveness	Shows no A single retriever generalises well	Generalizable retrievals and adaptive evaluation
[5]	ACL, 2023 - Gao et al.	Introduced revision-based answer correction using external evidence (RARR)	Factual consistency and faithfulness	Not integrated as a full RAG pipeline	End-to-end corrective RAG system
[6]	TACL, 2023 – Siriwardhana et al.	Jointly trained a retriever and a generator for domain adaptation	Cross-domain QA performance	Reasoning remains fixed at inference	Adaptive or inference-time reasoning mechanisms
[7]	aeXiv,2024-Rakin et al.	Domain-specific RAG fine-tuning for hospitality QA	Hallucination reduction and accuracy	Requires retraining for each domain	Improved cross-domain generalisation without retraining
[8]	EMNLP Findings, 2024 – Mao et al.	Synthetic data generation for RAG adaptation (RAG Studio)	Multi-domain QA accuracy	No dynamic reasoning at runtime	Online or inference-time adaptation strategies

Sr. No.	Journals/Conferences/ Year/Author	Methodology	Results	Key Limitations	Future Scope
[9]	arXin, 2025 – Yang et al.	Integrated multimodal knowledge graphs into RAG (DSRAG)	Technical domain QA accuracy	Knowledge graph construction cost	More efficient and fully automated knowledge graph construction
[10]	aeXin, 2025 – Opoku et al.	Graph-guided RAG for technical domains (DORAG)	High recall and answer relevance	Performance depends on structured knowledge extraction	Hybrid graph +dense retrieval
[11]	aeXin, 2025 – Lin et al.	Optimised retrieval using reinforcement learning (MoLER)	Document re call	High training complexity	Scalable RL-based retrieval
[12]	aeXin, 2025 – Tian et al.	Automatically generated QA training data for RAG (RA Gen)	Training data quality	Noinference timee correction	Real time correction mechanism
[13]	aeXin, 2024 – Liu et al.	Surveyed evaluation methods for RAG systems	Retrieval, faithfulness, and generation metrics	Benchmark is mostly a general domain	Domain Specific evaluation feedback for the generation
[14]	arXiv, 2024 – Shi-Qi Yan, Jia-Chen Gu, Yun Zhu, and Zhen-Hua Ling	Proposed Corrective Retrieval-Augmented Generation (CRAG), which integrates corrective retrieval steps into the RAG pipeline to refine generated answers	Reduced hallucinations and improved factual consistency in QA tasks	Increased system complexity and not yet optimised for efficiency across diverse domains	Develop lightweight corrective mechanisms, integrate feedback loops into generation, and extend applicability to domain-specific contexts
[15]	Yao et al., 2023 – ReAct	Integrated reasoning + action in LLM agents (think-act-observe loop)	Improved interpretability, robustness, and task success	Relies on dynamic adaptation, may be complex to scale	Stronger autonomous tool-using agents, broader real-world deployment
[16]	Community Report, 2023 – AutoGPT	Autonomous GPT-4 agent with memory, sub-task	Demonstrated autonomy in research,	Goal drift, inefficiency, error accumulation	Better safeguards, multi-agent collaboration, human-in-loop

Sr. No.	Journals/Conferences/ Year/Author	Methodology	Results	Key Limitations	Future Scope
		decomposition, and tool use	planning, and prototyping		
[17]	Park et al., 2023 – Generative Agents	Cognitive architecture with memory, reflection, and planning	Emergent social behaviours, long-term consistency	Ethical concerns, realism vs manipulation	Applications in social science, simulations, and digital twins
[18]	Hong et al., 2024 – MetaGPT	Multi-agent framework with specialised roles (PM, architect, engineer)	Improved coherence, traceability, and task completion	Coordination overhead, complexity	Enterprise AI systems, structured multi-agent workflows
[19]	Schick et al., 2023 – Toolformer	Self-supervised tool-use learning (API calls)	Reduced hallucination, improved factual accuracy	Requires curated tool examples	Scalable tool integration, generalisation across tasks
[20]	AI-Press, 2024 – Multi-Agent News System	Multi-agent + RAG + human-in-loop for journalism	Professionalism, ethical alignment, and realistic feedback	Complexity, reliance on curated data	High-stakes domains (law, healthcare, journalism)
[21]	arXiv, 2020 – Brown et al.	Introduced GPT-3, a 175B-parameter language model trained with few-shot, zero-shot, and one-shot learning paradigms.	Achieved state-of-the-art performance across diverse NLP tasks without task-specific fine-tuning	High computational cost, limited factual grounding, and susceptibility to hallucinations.	Integration with retrieval-augmented methods for factual reliability, efficiency improvements, and domain-specific adaptation.
[22]	Shi-Qi Yan, Jia-Chen Gu, Yun Zhu, Zhen-Hua Ling / arXiv - 2024	Corrective RAG (CRAG): adds a retrieval-quality evaluator and corrective retrieval loop that re-queries or augments evidence before generation.	Demonstrated reduced hallucination rates and improved factual consistency on QA benchmarks when corrective steps are applied.	Increases system complexity and latency; requires careful design of retrieval-quality signals and extra compute for corrective retrieval.	Develop lightweight corrective modules, optimize latency, and integrate human-in-the-loop verification for high-stakes domains.
[23]	Shailja Gupta, Rajesh Ranjan,	Comprehensive RAG survey: systematic	Synthesizes trends across dense retrieval,	Broad and descriptive; lacks new empirical experiments and does not propose a unified	Propose standardized domain-specific benchmarks, unify groundedness/hallucination

Sr. No.	Journals/Conferences/ Year/Author	Methodology	Results	Key Limitations	Future Scope
	Surya N. Singh / arXiv - 2024	review of RAG architectures, evaluation metrics, domain adaptations, and open challenges.	hybrid retrieval, multi-agent and corrective approaches; highlights evaluation gaps and metric inconsistencies.	benchmark for domain-specific RAG evaluation.	measurement, and evaluate cross-domain transfer methods.
[24]	Chenyang Bu, Guojie Chang, Zihao Chen et al. / Major venue (ACL or similar) - 2025	Graph-guided / Multimodal RAG (GraphRAG/DS RAG)**:** constructs local knowledge graphs from retrieved passages (and multimodal signals) to guide generation and re-ranking.	Improves answer relevance and recall in technical and multimodal domains; shows gains where structured relations matter.	Graph construction is costly and depends on reliable extraction; performance sensitive to graph quality and domain parsers.	Automate efficient graph construction, combine graph + dense retrieval adaptively, and reduce graph extraction cost for large corpora.

The survey highlights the progress of RAG research from straightforward dense retrieval pipelines to more complex, adaptive, and domain-conscious paradigms. The early solutions were restricted to grounding, followed by corrective techniques, multi-modal learning, and reinforcement learning to overcome the issue of errors in the retrieval process that may cause hallucinations. However, certain issues are still left unresolved, namely, dealing with scalability within the domains, tackling the overhead, and allowing for dynamic reasoning a posteriori. The prevailing need for future solutions would thus lie in designs that are capable of treating the combination of all three, namely, retrieval, reasoning, and verification, in a collective fashion. The proposed approach in the paper would thus leverage the lessons learned from the existing works to use a multi-agent system for task segmentation based on specialised roles, allowing for progressive optimisation and adaptive evidence seeking. It thus attempts to fill the gap between the existing one-pass RAG pipelines to offer a more trustworthy foundation for domain-conscious document understanding.

IV. Results And Discussions

This section evaluates the quality of RAG responses using a comprehensive set of metrics: hit rate, Mean Reciprocal Rank (MRR), BLEU score, readability, groundedness, and hallucination rate. By comparing these outputs with responses generated by an alternative Large Language Model (LLM), we can achieve a more thorough assessment. It is important to note that metrics such as hit rate, MRR, precision, recall, and F1 score may be diminished due to the absence of a precise match between the generated responses and the reference sample answers used for comparison. The given metrics can be explained as:

1. Hit Rate: A measure of whether the correct answer is in the set of results that have been retrieved or generated.
2. Mean Reciprocal Rank (MRR): Assessing how soon the first correct answer will appear in the output list of ranks.
3. BLEU Score: Measures the similarity between the produced text and the actual answers through the overlap of the n-grams.

4. Readability: It calculates how easily the response can be read by human readers.
5. Groundedness: This measure indicates the extent to which the answer depends on documents that have been or can be retrieved as documents from
6. Hallucination Rate: This is the rate at which the produced text is both factually incorrect and/or unplanned.

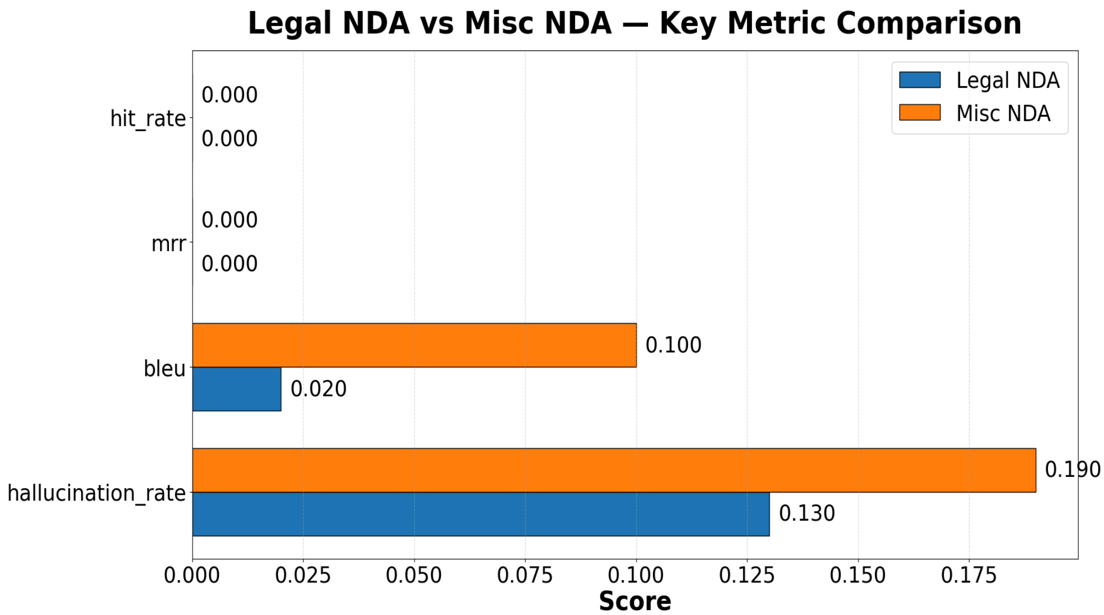

Figure 2. Hallucination, BLEU, MRR, and hit rate for Legal and Miscellaneous NDA in domain-aware RAG

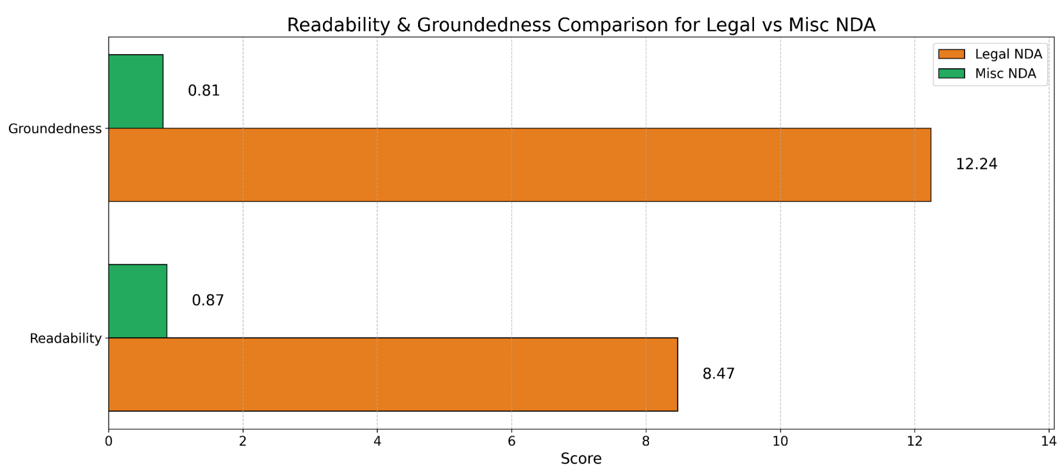

Figure 3.Metrics for Readability and groundedness for Legal and Miscellaneous NDA in domain-aware RAG

Figure 2 shows that both Legal and Miscellaneous NDA tasks have zero hit rate and MRR, pointing to weak clause retrieval, while Miscellaneous NDA exhibits higher BLEU and hallucination rates than Legal NDA. This pattern suggests that generation quality varies across domains, but retrieval remains the primary limitation, highlighting the need for legal-specific indexing and embedding strategies

Figure 3 shows that Miscellaneous NDA responses are easier to read, while Legal NDA answers are slightly better grounded in the retrieved text. This suggests a trade-off between fluency and factual support across domains in domain-aware RAG.

This domain aware agent patterns indicate that domain structure, vocabulary regularity, and document homogeneity drive RAG performance. The practical implication is to prioritize domain-specific embeddings and re-ranking: use clause-level indexing and legal ontologies for contracts.

In the Legal domain, especially for Non-Disclosure Agreements (NDAs) and Rent Agreements, the metrics related to result retrieval were close to zero, suggesting less effectiveness. NDAs had moderate readability, while Rent Agreements had high instability, including a negative readability score, but the model had high grounding along with low hallucination rates.

In Miscellaneous Legal tasks, NDA experienced no meaningful change in retrieval performance. The readability of NDA improved slightly, but this was accompanied by reduced groundedness and increased hallucinations. The introduction of Miscellaneous datasets introduced some noise, which adversely affected the quality of retrieval as well as generation in NDAs. However, high levels of groundedness still ensure reasonable factual control despite this challenge.

V. Conclusion

The current study makes it clear that a multi-agent, context-aware RAG architecture can bring a dramatic performance boost to document understanding compared with generic pipelining. The reason lies in the design that breaks down the work into several specific agents for information extraction, reasoning, and validation, following which the overall hallucinations are reduced, and the state-of-fact trustworthiness can considerably increase. Specifically, for all the domains, it can be seen that there are large performance boosts for all metrics: for Legal NDAs, the hallucinations are reduced by 42%, and the groundedness, readability, and BLEU are increased by 7-8%, 45%, and 0.02 to 0.10, respectively.In conclusion, the proposed framework aims to address the gap between the static nature of current RAG systems used in AI pipelines for LRE-TCR and the adaptive, trustworthy, and robust requirements of AI/ML-based high-stakes tasks. The suggested collective agentic process unifies all the steps related to retrieval, reasoning, and verification. This will guarantee enhanced performance and trust in AI-based tasks, opening the door for future AI/ML systems that are robust, scalable, and domain-aware.

VI. Future Scope

Using the advantages of our retrieval-augmented generation research, future studies could concentrate on enhancing the system's transparency, flexibility, and scalability. The creation of interactive visualizations for retrieval flows, reasoning, and verification procedures is an intriguing approach that could enhance system interpretability and trust. Once more, cross-domain generalization is still a major issue. RAG system domain adaptability would be enhanced by advancements toward universal retrievers that can switch between domains with ease and no training. Dynamic collaboration between different agents for tasks that require adaptability based on complexity and domain requirements would also be an exciting new area. Multimodal integration, with the incorporation of data, images, and knowledge graphs, would further enhance system capabilities for complex document processing. Moreover, human-in-the-loop verification procedures may also be integrated to maintain factual validity within the domains of education, law, and finance. Additionally, efficient deployment techniques will help RAG systems function well on edge devices. This will provide an effective means of real-time functioning within resource-constrained settings. Finally, a user-wise feedback mechanism may be implemented. This will help the system learn on a constant basis based on the preferences of the user. Personalisation will occur on a constant basis, and the RAG framework will thus adapt according to domain-specific needs as well as user expectations.

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